QIANG ZHU

Department of Physics and Astronomy, University of Nevada Las Vegas, Las Vegas, NV 89154-4002
Phone: (702)-895-1707 Email: qiang.zhu@unlv.edu Web: http://www.physics.unlv.edu/ qzhu

EDUCATION

Ph.D., Geosciences, *State University of New York at Stony Brook* (2009-2013)

B.S., Materials Science and Engineering, *Beihang University* (2003-2007)

EMPLOYMENT

Assistant Professor, University of Nevada, Las Vegas, Oct 2016 - Present

Research Assistant Professor, Stony Brook University, Feb 2014 - Sept 2016

Research Associate, Stony Brook University, Feb 2013 - Feb 2014

- Chief developer of the USPEX code for crystal structure prediction ($\sim$ 2500 users).
- Organic crystal polymorphism
- Materials defects and under extreme conditions
- Search and design of (in)organic materials with superior properties

CAREER RECORD

- 40 papers in peer reviewed journals, including 1 in Science, 2 in Nature Chemistry, 2 in Nature Communicatioin, 5 in PRL, 2 in Angew. Chem., 1 in JACS, 1600+ citations, H index 20
- 30+ invited talks
- Lecturer in a series of international worshops (France, 2011/2015; China, 2011/2013/2015/2017; Switzerland, 2012; USA, 2012; Isarel, 2013; Canada, 2014; India, 2015; Italy, 2016)
- 2 patents
- Editorial Member of Scientific Reports since 2015
- Reviewer for *Nature Comm., J. Am. Chem. Soc., Scientific Reports, App. Phys. Lett., Phys. Chem. Chem. Phys., Euro. Phys. Lett., Solid State Comm., J. Mater. Chem. C, J. Non-Crystalline Solids, Chem. Mater., Acta Cryst*
- President's Award for Distinguished Doctoral Students, Stony Brook University, 2013

SELECTED PUBLICATIONS

1. **Q. Zhu**, *et al.*, Predicting phase behavior of grain boundaries with evolutionary search and machine learning. Nature Communication, Accepted, 2018
2. A.G. Shtukenberg, **Q. Zhu***, *et al.*, Powder diffraction and crystal structure prediction identify four new coumarin polymorphs. Chemical Science, 8(2017), 4926-4940
3. W. Xu, **Q. Zhu***, Hu CT, Structure of Glycine Dihydrate: Its implications to Crystallization of Glycine from Solution and Modification of Glycine in Space. Angew. Chem. Int. Ed., 129(2017), 2062-2066
4. **Q. Zhu**, *et al.*, Resorcinol Crystallization from the Melt: A New Ambient Phase and New Riddles. J. Am. Chem. Soc., 138(2016), 4881-4889
5. X. Meng, L. Wang, D. Liu, X. Wen **Q. Zhu***, W. A. Goddad, Q. An, Discovery of Fe2P-Type Ti(Zr/Hf)2O6 Photocatalysts toward Water Splitting. Chem. Mater., 28(2016), 1335
6. **Q. Zhu***, *et al.*, Stability of xenon oxides at high pressures. Nature Chemistry, 5(2013), 61-65
7. W. Zhang, A. R. Oganov, A. Goncharov, **Q. Zhu**, *et al.*, Unexpected stable stoichiometries of sodium chlorides. Science, 342(2013), 1502-1505